

29. Which of the following statements is true?

- A α particles can be stopped by 1 m of lead.
- B β particles can be deflected when travelling parallel to a magnetic field.
- C γ particles can be stopped by paper.
- D α , β and γ particles are all charged particles.

Ans: A

Option A: if α particles can be stopped by a thin piece of paper, it can surely be stopped by 1 m of lead.

Option B: β particles, which are charged particles, travelling parallel to a magnetic field will not experience a magnetic force. They need to cut across the magnetic field to experience a magnetic force.

Option C: γ particles (or rays) can only be stopped by about 10 cm of lead. They can't be stopped by paper.

Option D: γ particles (or rays) are uncharged electromagnetic radiation.

30. Technetium-99 has a half-life of 6.0 hrs. It is used as a biological tracer. A sample of